

Dennis Fang

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EDUCATION

University of California, San Diego

Sept 2025 - Mar 2027

Master of Science, Data Science

University of California, Davis

Sept 2021 - Jun 2025

Bachelor of Science, Statistics (Data Science Track), Atmospheric Sciences - GPA - 3.7/4.0

- **Coursework:** Multivariate Statistics, Machine Learning, Applied Linear Algebra, Advanced Statistical Computing, Statistical Learning, Practice in Statistical Data Science, Probability Theory, Mathematical Statistics, Time Series

RELEVANT SKILLS

- **Technical Skills:** Python, R, SQL, Tableau, Power BI, Azure, GitHub, Machine Learning, Data Visualization
- **Communication & Tools:** Microsoft Office Suite (Word, Excel, PowerPoint), Microsoft Access

WORK EXPERIENCE

Navy Research Laboratory | Monterey, CA

Jun 2024 - Aug 2024

Research Intern - Data Analysis

- Analyzed 100 km² of observational climate data close to Western Africa using Python (Pandas, NumPy), comparing WRF and NAAPS outputs to find differences in patterns of temperature and aerosol concentration and guide model calibration.
- Quantified a 31% improvement in the accuracy of meteorological and aerosol forecasts in NAAPS-RA v2 vs. NAAPS-RA v1 by applying RMSE metrics and time-series evaluations, providing significant insights into climate model differences.
- Processed, cleaned, and transformed 30+ GB of raw and gridded data using Python and Excel, and visualized critical findings derived from the cleaned dataset to highlight model performance improvements.
- Delivered insights through Quad Chart reports presented to project stakeholders.

Lawrence Berkeley National Lab | Berkeley, CA

Jun 2023 - Aug 2023

Research Intern - Data Analysis

- Collected and quality-checked over 300,000 environmental sensor readings (temperature, snow depth) from the East River Valley, Colorado, using Excel; organized the resulting dataset to facilitate further geospatial and climate model analysis.
- Evaluated relationships between solar radiation, topographic indices, and snowpack metrics using Python (Pandas, NumPy, Seaborn), and achieved an 87% classification model accuracy in predicting snowpack variability across high-altitude terrain.
- Co-authored a research manuscript and presented findings in an LBNL-hosted webinar, and used data visualizations to highlight important drivers of snowpack variability to 10+ researchers, facilitating future research collaboration.

ACADEMIC PROJECTS

Machine Learning Final Project - An ML Approach to Rainfall Prediction | Davis, CA

Nov 2024 - Dec 2024

- Built machine learning models (Random Forest, XGBoost, Time Series) to forecast rainfall patterns across 100 monitoring sites in Northern California using 5 years of historical weather data.
- Cleaned and organized a multi-year time series dataset, engineered features (lag variables, precipitation trends), and trained models with Pytorch, TensorFlow, and scikit-learn.
- Assessed ML model performance using MAE and R² scores for clusters of sensors using K-Means Clustering, and found that the Random Forest model outperformed time series and XGBoost, improving prediction accuracy by 47%.
- Documented project results and findings in an organized GitHub repository that includes visualizations, model evaluations, and written analyses of each model's advantages and disadvantages.

Data Analyst Final Project - Baseball Hall of Fame Classification | Davis, CA

May 2024 - Jun 2024

- Developed predictive models to estimate Baseball Hall-of-Fame induction likelihood using a dataset of 1,000 MLB players' career statistics, including home runs scored, batting average attained, and MVP awards won.
- Applied Logistic Regression, Linear Discriminant Analysis, and Decision Tree models using Python (Pandas, NumPy, matplotlib, scikit-learn), increasing prediction accuracy by 11% over the baseline model.
- Collaborated in a team of four to compile findings into a formal report and present the results to the course instructors in a final presentation, showcasing data visualizations and their relevance to finding key drivers of Hall-of-Fame induction rates.